

Configuring the Windows Defender Firewall (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 04

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Time on Task:

3 hours, 8 minutes

Progress:

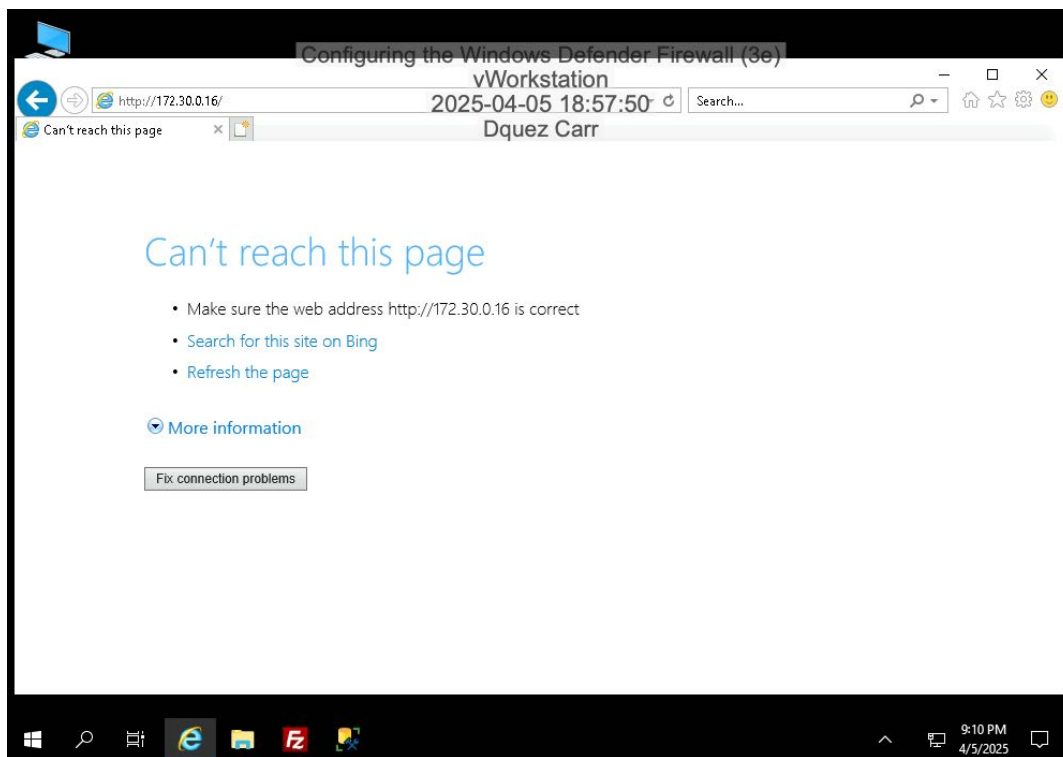
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Report Generated: Sunday, April 6, 2025 at 1:42 AM

Section 1: Hands-On Demonstration

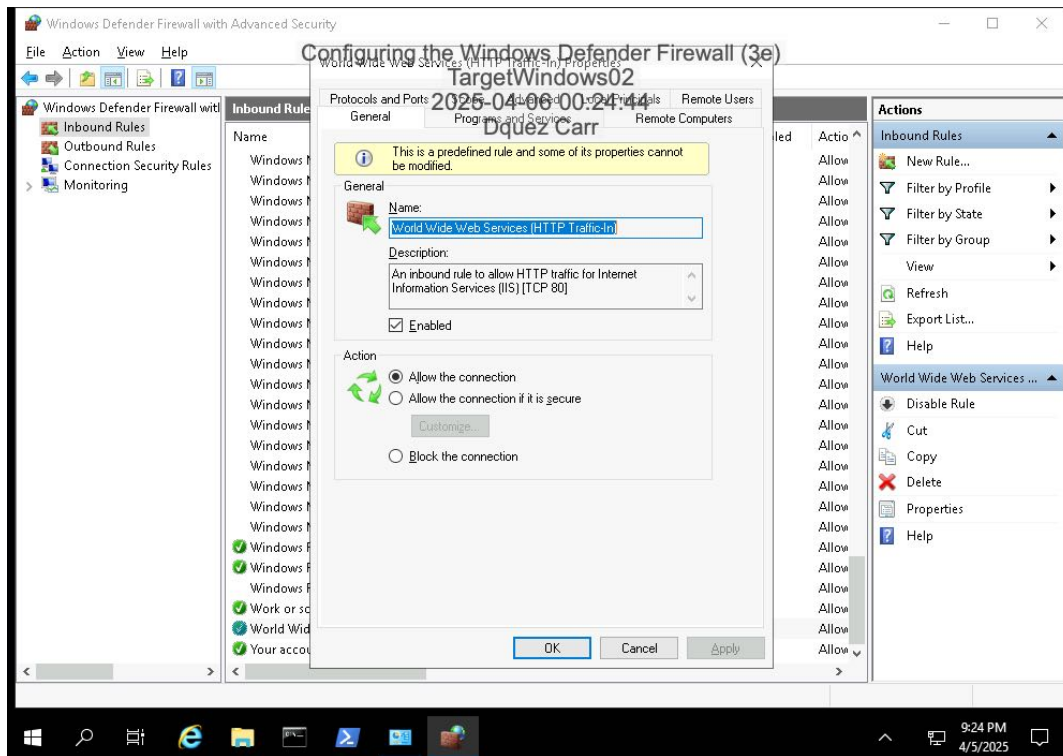
Part 1: Test Access to a Windows IIS Server Using a Web Browser

3. Make a screen capture showing the connection error message in the browser window.



Part 2: Enable a Windows Defender Firewall Rule to Allow HTTP Access

12. Make a screen capture showing the **World Wide Web Services (HTTP Traffic-In)** Properties dialog box and the **Enabled** checkbox.

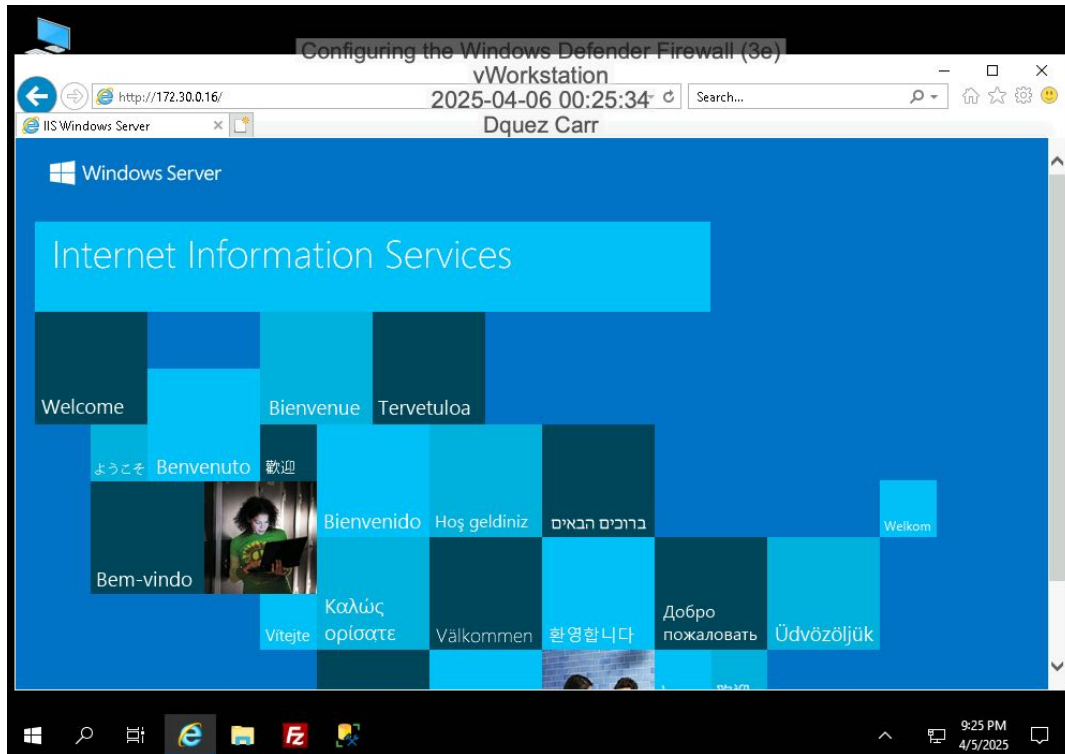


Part 3: Confirm the Windows Defender Firewall Configuration Allows HTTP

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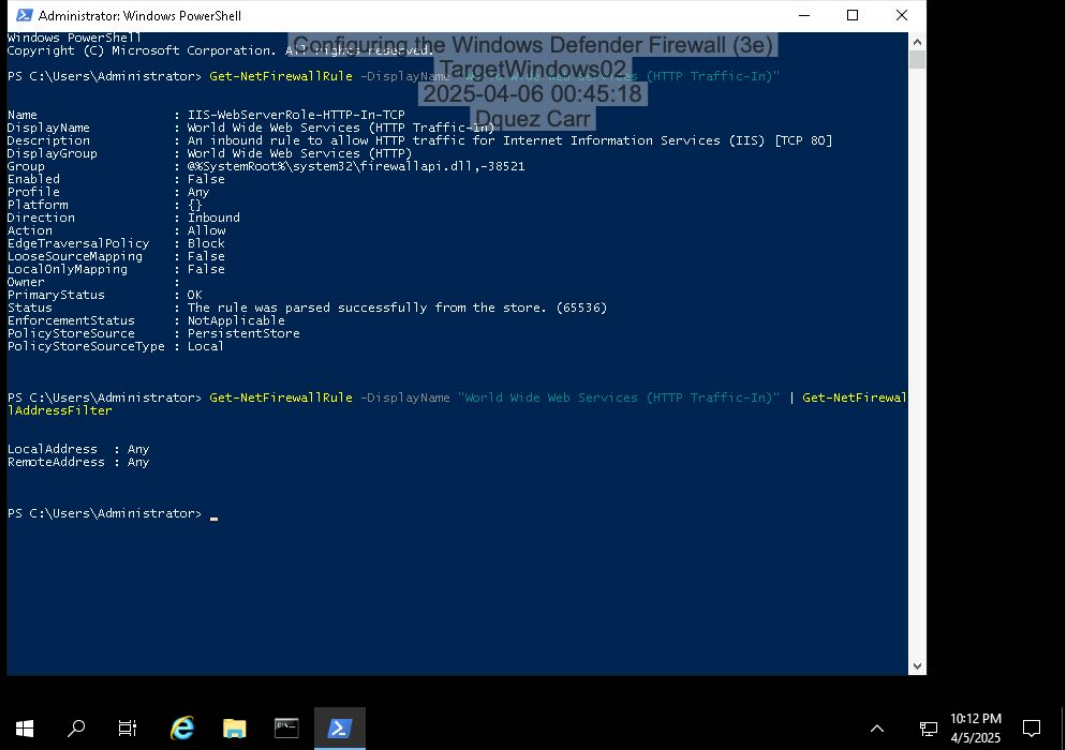
4. Make a screen capture showing the IIS server home page in the browser window.



Section 2: Applied Learning

Part 1: Test Access to Windows IIS Server Using PowerShell

6. Make a screen capture showing the results of the **Get-NetFirewallRule** commands.



The screenshot shows a Windows PowerShell window titled "Administrator: Windows PowerShell". The command executed is `Get-NetFirewallRule -DisplayName "World Wide Web Services (HTTP Traffic-In)"`. The output displays the properties of the firewall rule, including its name, display name, description, group, and status. The rule is named "IIS-WebServerRole-HTTP-In-TCP" and is part of the "World Wide Web Services (HTTP)" group. It is currently disabled. The command is then piped to `Get-NetFirewallRule -AddressFilter`, which shows the local and remote addresses as "Any".

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> Get-NetFirewallRule -DisplayName "World Wide Web Services (HTTP Traffic-In)"

Name                : IIS-WebServerRole-HTTP-In-TCP
DisplayName          : World Wide Web Services (HTTP Traffic-In)
Description          : An inbound rule to allow HTTP traffic for Internet Information Services (IIS) [TCP 80]
DisplayGroup         : World Wide Web Services (HTTP)
Group                : @%SystemRoot%\system32\firewallapi.dll,-38521
Enabled              : False
Profile              : Any
Platform             : {}
Direction           : Inbound
Action               : Allow
EdgeTraversalPolicy  : Block
LooseSourceMapping   : False
LocalOnlyMapping     : False
Owner                :
PrimaryStatus        : OK
Status               : The rule was parsed successfully from the store. (65536)
EnforcementStatus    : NotApplicable
PolicyStoreSource     : PersistentStore
PolicyStoreSourceType : Local

PS C:\Users\Administrator> Get-NetFirewallRule -DisplayName "World Wide Web Services (HTTP Traffic-In)" | Get-NetFirewallRule -AddressFilter

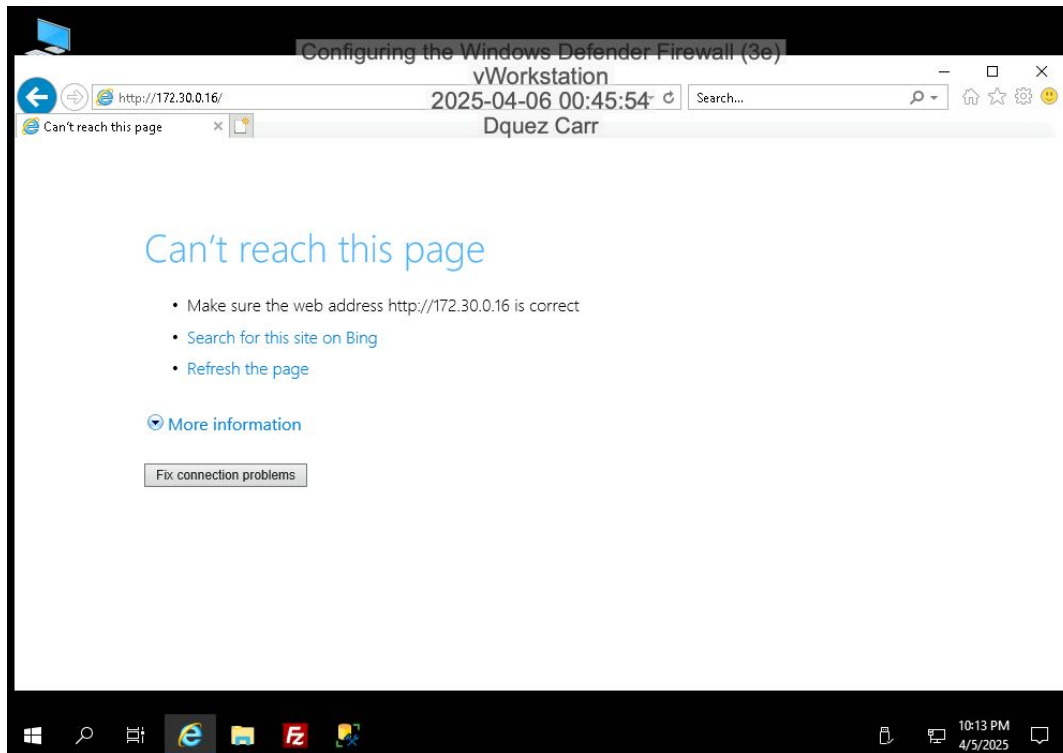
LocalAddress : Any
RemoteAddress : Any

PS C:\Users\Administrator>
```

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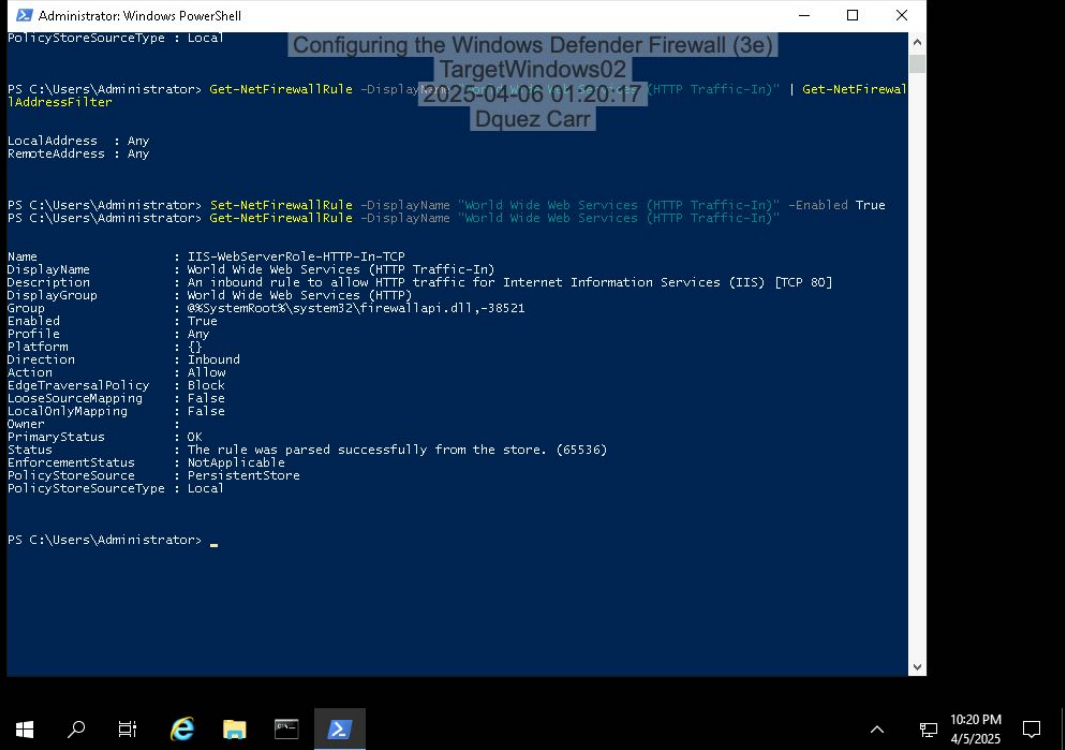
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9. Make a screen capture showing the connection error message in the browser window.



Part 2: Enable a Windows Defender Firewall Rule to Allow HTTP Access

4. Make a screen capture showing the updated status of the World Wide Web Services rule.



```
Administrator: Windows PowerShell
PolicyStoreSourceType : Local

PS C:\Users\Administrator> Get-NetFirewallRule -Display (HTTP Traffic-In) | Get-NetFirewallRule -AddressFilter

LocalAddress : Any
RemoteAddress : Any

PS C:\Users\Administrator> Set-NetFirewallRule -DisplayName "World Wide Web Services (HTTP Traffic-In)" -Enabled True
PS C:\Users\Administrator> Get-NetFirewallRule -DisplayName "World Wide Web Services (HTTP Traffic-In)"

Name                : IIS-WebServerRole-HTTP-In-TCP
DisplayName          : World Wide Web Services (HTTP Traffic-In)
Description          : An inbound rule to allow HTTP traffic for Internet Information Services (IIS) [TCP 80]
DisplayGroup         : World Wide Web Services (HTTP)
Group                : @%SystemRoot%\system32\firewallapi.dll,-38521
Enabled              : True
Profile              : Any
Platform             : {}
Direction            : Inbound
Action               : Allow
EdgeTraversalPolicy  : Block
LooseSourceMapping   : False
LocalOnlyMapping     : False
Owner                :
PrimaryStatus        : OK
Status               : The rule was parsed successfully from the store. (65536)
EnforcementStatus    : NotApplicable
PolicyStoreSource    : PersistentStore
PolicyStoreSourceType : Local

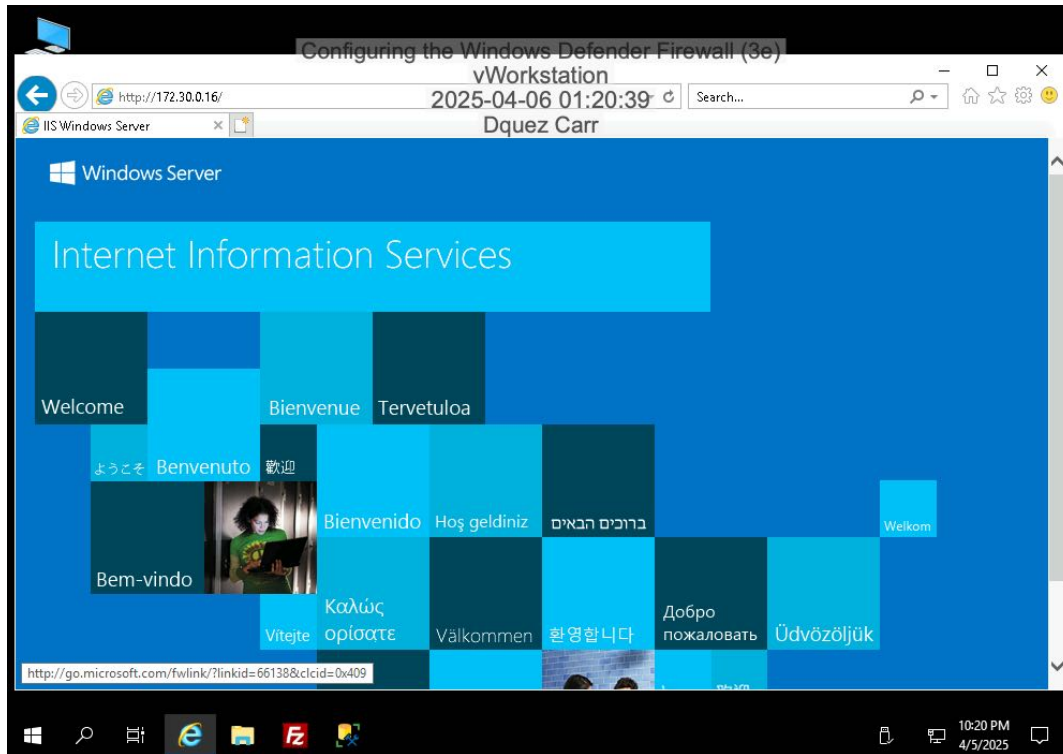
PS C:\Users\Administrator>
```

Part 3: Confirm the Windows Defender Firewall Configuration Allows HTTP

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3. Make a screen capture showing the IIS server home page in the browser window.



Section 3: Challenge and Analysis

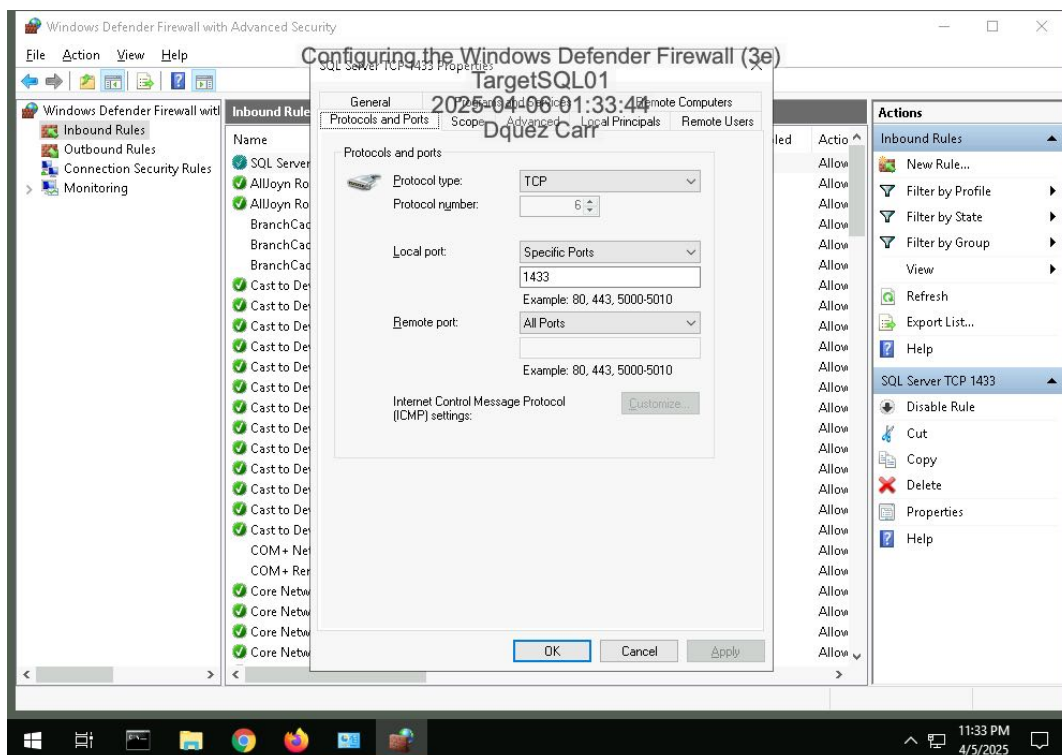
Part 1: Research Firewall Rule Requirements

Using the Internet, **research** and **identify** the protocol and port necessary for a Microsoft SQL Server database connection.

By default, Microsoft SQL Server connects to databases through port 1433 using the TCP protocol. This port needs to be open on all firewalls and network devices that connect the client and server in order for communication to be successful.

Part 2: Configure the Firewall

Make a screen capture showing your firewall rule on TargetSQL01.



Part 3: Validate Your Firewall Rule

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Make a screen capture showing the results of your SQL query.

The screenshot shows the Microsoft SQL Server Management Studio (SSMS) interface. The title bar indicates the connection is to '172.30.0.17 (SQL Server 14.0.1000 - bross (53))'. The 'Object Explorer' on the left shows the server structure. The 'Query Editor' in the center contains a SQL query: `SELECT * FROM countries;`. The 'Results' pane displays a table with 18 rows of country data. The status bar at the bottom confirms the query was executed successfully, returning 227 rows.

	Country	Region	Population	Area_sq_mi	Pop_Density_per_sq_mi	Coastline_coast_area_ratio	N
1	Afghanistan	ASIA (EX. NEAR EAST)	31056997	647500	480	0	2
2	Albania	EASTERN EUROPE	3581655	28748	1246	126	-4
3	Algeria	NORTHERN AFRICA	32930091	2381740	138	4	-1
4	American Samoa	OCEANIA	57794	199	2904	5829	-2
5	Andorra	WESTERN EUROPE	71201	468	1521	0	6
6	Angola	SUB-SAHARAN AFRICA	12127071	1246700	97	13	0
7	Anguilla	LATIN AMER. & CARIB	13477	102	1321	5980	1
8	Antigua & Barbuda	LATIN AMER. & CARIB	69108	443	1560	3454	-1
9	Argentina	LATIN AMER. & CARIB	39921833	2766890	144	18	0
10	Armenia	C.W. OF IND. STATES	2976372	29800	999	0	-1
11	Aruba	LATIN AMER. & CARIB	71891	193	3725	3549	0
12	Australia	OCEANIA	20264082	7686850	26	34	3
13	Austria	WESTERN EUROPE	8192880	83870	977	0	2
14	Azerbaijan	C.W. OF IND. STATES	7961619	86600	919	0	-4
15	Bahamas, The	LATIN AMER. & CARIB	303770	13940	218	2541	-2
16	Bahrain	NEAR EAST	698585	665	10505	2421	1
17	Bangladesh	ASIA (EX. NEAR EAST)	1473653...	144000	10234	40	-1
18	Barbados	LATIN AMER. & CARIB	279312	431	6495	2251	-1